

ROAD-SEGMENT RISK CLASSIFICATION USING DECISION TREES AND LIGHTGBM

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Introduction

Road Safety & Accident Risk

Road accidents are a major concern worldwide.

Accidents can cause:

- Loss of lives

- Serious injuries

- Property damage

Some road segments experience accidents repeatedly.

Identifying high-risk road segments in advance can help authorities take preventive action.

Machine Learning can be used to classify roads according to their accident risk.

Problem Statement

It is difficult to manually identify accident-prone road segments.

Different factors can contribute to road accident risk.

Authorities need a data-driven method to identify risky locations.

The system should classify road segments into different **risk levels**.

Accurate classification can help authorities prioritize safety improvements.

Objectives of the Project

Analyze road-related data using Machine Learning.

Identify factors associated with road accident risk.

Preprocess and prepare the dataset for analysis.

Build a **Decision Tree** classification model.

Build a **LightGBM** classification model.

Compare both models using evaluation metrics.

Identify the better-performing model.

Support authorities in making road-safety decisions.

Dataset & Features

Input Features

The proposed system uses information such as:

Traffic Density

Weather Conditions

Speed Limit

Road Geometry

Accident History

Road Surface Conditions

Output:

Classification of a road segment into different **risk categories**.

Data Preprocessing

Before training the models, the dataset is prepared through:

1. Handling Missing Values

Identify missing information.

Fill or handle missing values appropriately.

2. Removing Unnecessary Data

Remove irrelevant or redundant information.

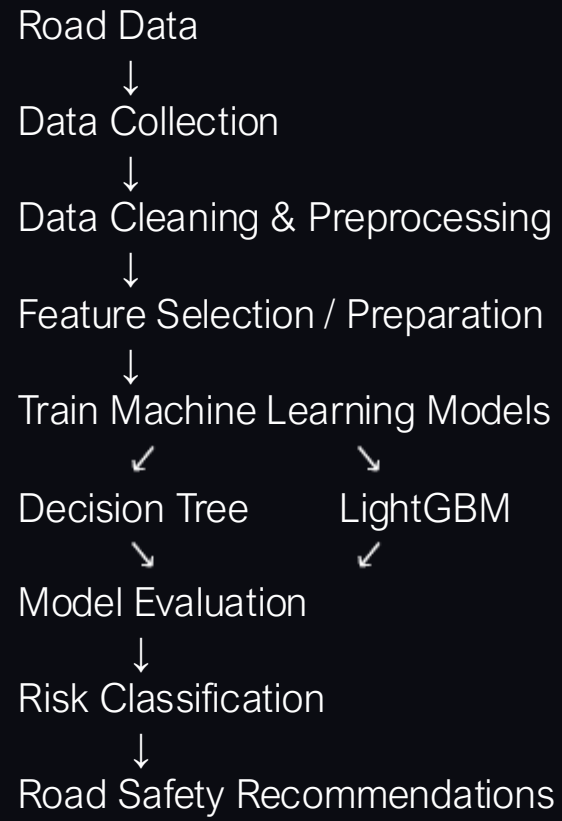
3. Data Transformation

Convert the data into a suitable format for Machine Learning.

4. Prepared Dataset

Use the cleaned dataset for model training and testing.

Proposed Methodology



Decision Tree Algorithm

A Decision Tree is a classification algorithm.

It makes predictions through a series of decision rules.

It is easy to understand and interpret.

It clearly shows how different factors influence the prediction.

It can classify road segments based on their characteristics.

Example:

Traffic Density → High?

↓

Weather → Poor?

↓

Accident History → High?

↓

Higher Risk Category

LightGBM Algorithm

LightGBM is a machine-learning algorithm suitable for classification.

It is designed to work efficiently with large datasets.

It can handle complex patterns in data.

It provides high prediction accuracy.

In this project, LightGBM is compared with the Decision Tree model.

Why LightGBM?

Faster

Efficient

Handles large datasets

Better at complex patterns

High prediction accuracy

Performance Evaluation

Metric	Purpose
Accuracy	Overall correctness of predictions
Precision	Correctness of predicted risk categories
Recall	Ability to identify relevant risky segments
F1-Score	Balance between precision and recall

The two models are compared using:

Models Compared:

- Decision Tree
- LightGBM

Results & Findings

Results

According to the abstract:

Decision Tree

- Easier to interpret

- Provides a clear classification process

LightGBM

- Provides better prediction accuracy

- Handles complex data more effectively

Overall Finding

LightGBM performs better than Decision Tree for this project.

Applications & Future Impact

Applications

The developed system can help:

Traffic authorities

Road planners

Government agencies

Possible Safety Measures

Improve road maintenance

Install warning signs

Implement speed-control systems

Prioritize accident-prone areas

Improve road planning

Software Requirements

Requirement	Technology
Operating System	Windows 10/11
Programming Language	Python 3.x
IDE	Jupyter Notebook / VS Code
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Dataset	CSV
Browser	Google Chrome / Microsoft Edge

Conclusion

Machine Learning can help identify high-risk road segments.

The project uses **Decision Tree and LightGBM** for risk classification.

Road-related factors such as traffic, weather, speed limits, road geometry, accident history, and surface conditions are considered.

Decision Tree provides better interpretability.

LightGBM provides better prediction accuracy and handles complex data more effectively.

The system can support authorities in taking preventive road-safety measures.

Ultimately, the project aims to contribute to **safer roads and fewer accidents.**